Trigonometry – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The problems are ordered by difficulty. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 1/5 (easy)*

Let F_n and L_n be the Fibonacci and Lucas numbers ($F_0 = 0$, $F_1 = 1$, $L_0 = 2$, $L_1 = 1$, and $x_{n+1} = x_n + x_{n-1}$ for both). Prove that for every positive integer n

$$\sum_{k=0}^{n-1} \frac{1}{3 - 2\cos\frac{2\pi k}{n}} = \begin{cases} \dfrac{nL_n}{5F_n}, & n \text{ even}, \\ \dfrac{nF_n}{L_n}, & n \text{ odd}. \end{cases}$$

Based on: L. Smajlović, Z. Šabanac, L. Šćeta, *Relations between Chebyshev, Fibonacci and Lucas polynomials via trigonometric sums*, arXiv:2403.12516, Corollary 5.3.

Problem 2. *IMC level: 1/5 (easy)*

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $f(t) = \cos t + 2\cos(\varphi t) + 2\cos(\varphi^2 t) + \cos(\varphi^3 t)$ for $t \in \mathbb{R}$.

(a) Prove that $\min_{t\in\mathbb{R}} f(t) = -2$, $\max_{t\in\mathbb{R}} f(t) = 6$, and that $f(t) = 6$ only for $t = 0$.

(b) Prove that nevertheless $\sup_{t\geqslant T} f(t) = 6$ for every $T > 0$, and deduce that f is not periodic.

Based on: M. Przybylski, *The golden quadruple: four reals whose differences form a geometric progression, rigidity of the golden ratio, and a Bohr almost periodic cosine sum*, arXiv:2611.00973, Theorem 4, Propositions 6.2 and 6.5.

Problem 3. *IMC level: 1/5 (easy)*

For a positive integer N let c_N be the number of integers n with $1 \leqslant n \leqslant N$ and $\cos n > \cos 1$. Prove that

$$0 < \frac{N}{\pi} - c_N < 2 \qquad \text{for all } N \geqslant 1.$$

Based on: M. M. Etkind, S. Grepstad, M. N. Kolountzakis, N. Lev, *Bounded remainder sets, bounded distance equivalent cut-and-project sets, and equidecomposability*, arXiv:2511.21148, Section 3.2 and Theorem A.1 (Hecke's theorem on bounded remainder intervals).

Problem 4. *IMC level: 2/5 (below average)*

Find all pairs (x, y) of rational numbers with $0 \leqslant x \leqslant y \leqslant 1$ such that $\cos \pi x + \cos \pi y$ is rational.

Based on: Q. Xue, *On structured cosine sums and applications*, arXiv:2607.20907, Proposition 2.12.

Problem 5. *IMC level: 2/5 (below average)*

For $n \geqslant 1$ define polynomials $P_n, Q_n \in \mathbb{Z}[t]$ by $(1+it)^n = Q_n(t) + iP_n(t)$, so that $\tan n\theta = P_n(\tan\theta)/Q_n(\tan\theta)$. Let p be an odd prime.

(a) Show that $P_n(x)$ and $Q_n(x)$ never vanish simultaneously for $x \in \mathbb{F}_p$.

(b) Define $f_n\colon \mathbb{F}_p \cup \{\infty\} \to \mathbb{F}_p \cup \{\infty\}$ by $f_n(x) = P_n(x)/Q_n(x)$ if $Q_n(x) \neq 0$, $f_n(x) = \infty$ if $Q_n(x) = 0$, and $f_n(\infty) = \infty$ for odd n, $f_n(\infty) = 0$ for even n. Prove that f_n is a bijection if and only if $\gcd\bigl(n,\, p - (-1)^{(p-1)/2}\bigr) = 1$.

Based on: Z. Ding, M. E. Zieve, *Tangent-Chebyshev rational maps and Rédei functions*, arXiv:2103.08128, Corollaries 3.3 and 3.5.

Problem 6. *IMC level: 2/5 (below average)*

For smooth functions $g_1, \ldots, g_n$ let $\mathrm{Wr}(g_1, \ldots, g_n) = \det\bigl(g_j^{(i)}\bigr)_{0\leqslant i\leqslant n-1,\ 1\leqslant j\leqslant n}$ be their Wronskian.

(a) Prove that $\mathrm{Wr}(\sin x, \sin 2x, \ldots, \sin nx) = (-2)^{\frac{n(n-1)}{2}} \Bigl(\prod_{k=0}^{n-1} k!\Bigr) \sin^{\frac{n(n+1)}{2}} x$.

(b) Prove that $\mathrm{Wr}\bigl(\sin x, \ldots, \sin(n-1)x, \sin(n+1)x\bigr) = (-2)^{\frac{n(n-1)}{2}} \Bigl(\prod_{k=0}^{n-1} k!\Bigr) 2n\cos x\, \sin^{\frac{n(n+1)}{2}} x$.

Based on: M. Yuan, *On Gegenbauer polynomials and Wronskian determinants of trigonometric functions*, arXiv:2501.11092, formula (1), Theorem 1 and Proposition 1.

Problem 7. *IMC level: 3/5 (medium)*

For an odd integer d let $\chi(d) = (-1)^{(d-1)/2}$. Let $k \geqslant 3$ be odd. Prove that

$$\sum_{\substack{1\leqslant \ell < k,\ \ell \text{ odd} \\ \gcd(\ell,k)=1}} \frac{\chi(\ell)}{\sin\frac{\pi\ell}{2k}} = \frac{k}{2} \prod_{p\mid k} \Bigl(1 - \frac{\chi(p)}{p}\Bigr),$$

where the product is over the prime divisors p of k.

Based on: B. C. Berndt, S. Kim, A. Zaharescu, *Evaluations and relations for finite trigonometric sums*, arXiv:2403.03445, Theorems 2.1 and 3.2.

Problem 8. *IMC level: 3/5 (medium)*

Let A be a set of n positive integers which is *sum-free*: there are no $a, b, c \in A$ (not necessarily distinct) with $a + b = c$. Prove that

$$\min_{x\in\mathbb{R}} \sum_{a\in A} \cos(ax) \leqslant -\sqrt{n/2}.$$

Based on: B. Bedert, *Polynomial bounds for the Chowla cosine problem*, arXiv:2509.05260, Lemma 4.1.

Problem 9. *IMC level: 3/5 (medium)*

Prove that for all integers $n \geqslant 2$ and $m \geqslant 0$ the number

$$\left(\frac{n}{2}\right)^m \sum_{k=1}^{n-1} \frac{1}{\sin^{2m}(k\pi/n)}$$

is an integer.

Based on: J. Jorgenson, A. Karlsson, L. Smajlović, *The resolvent kernel on the discrete circle and twisted cosecant sums*, arXiv:2305.00202, Section 1 (integrality of the Verlinde numbers).

Problem 10. *IMC level: 3/5 (medium)*

Let $N \geqslant 1$ and let x be a real number with $\sin(nx) \neq 0$ for $n = 1, \ldots, N$. Prove that

$$\sum_{n=1}^{N} \frac{1}{|\sin nx|} > \frac{N}{\pi}(\ln N - 1).$$

Based on: B. Borda, *Remarks on sums of reciprocals of fractional parts*, arXiv:2305.07394, Introduction and Theorem 1.

Problem 11. *IMC level: 4/5 (above average)*

Find all rational numbers q for which there exist real numbers α, β, both rational multiples of π, such that $\tan\alpha$ and $\tan\beta$ are defined, $\tan\beta \neq 0$, and $\tan\alpha = q\tan\beta$.

Based on: Y. Liao, E. Wang, *Tilings of the sphere by congruent quadrilaterals II: edge combination a^3b with rational angles*, arXiv:2205.14936, Section 7.

Problem 12. *IMC level: 4/5 (above average)*

For pairwise coprime positive integers a, b, c put $C(a;b,c) = \sum_{k=1}^{a-1} \cot\frac{\pi kb}{a}\cot\frac{\pi kc}{a}$ (an empty sum is 0). Prove that

$$\frac{C(a;b,c)}{a} + \frac{C(b;c,a)}{b} + \frac{C(c;a,b)}{c} = \frac{a^2+b^2+c^2}{3abc} - 1.$$

Based on: B. C. Berndt, S. Kim, A. Zaharescu, *Evaluations and relations for finite trigonometric sums*, arXiv:2403.03445, Theorem 7.3 and Lemma 7.2.

Problem 13. *IMC level: 4/5 (above average)*

Let n be an odd positive integer. Determine all triples (a, b, c) of integers such that

$$\cos\frac{2\pi a}{n} + \cos\frac{2\pi b}{n} + \cos\frac{2\pi c}{n} = 0.$$

Based on: Q. Xue, *On structured cosine sums and applications*, arXiv:2607.20907, Theorem 3.5, Corollary 3.6 and Theorem 2.3.

Problem 14. *IMC level: 4/5 (above average)*

Let $a > 1$ and let $n \geqslant 2$ be an integer. Determine

$$\min_{p} \max_{-1 \leqslant x \leqslant 1} \frac{|p(x)|}{a - x},$$

where the minimum is taken over all monic real polynomials p of degree n.

Based on: O. Rubin, *Chebyshev polynomials in the complex plane and on the real line*, arXiv:2411.14175, Theorems 2.6 and 2.7 (Markov's weighted Chebyshev polynomials).

Problem 15. *IMC level: 5/5 (hard)*

For a positive integer n let

$$\sigma(n) = \min\{|z_1 + \cdots + z_n| : \ z_1^5 = \cdots = z_n^5 = 1, \ z_1 + \cdots + z_n \neq 0\}.$$

Prove that $\liminf_{n \to \infty} n\,\sigma(n) = \dfrac{1 + \sqrt{5}}{2}$.

Based on: A. Munemasa, G. Núñez Ponasso, *The minimal absolute value of sums of fifth roots of unity*, arXiv:2607.00825, Theorem 1.